

Day 18 - Lecture 18.12: Horizontal Bar Charts (`plt.barh`)

1. Overview & When to Use Horizontal Bars

When category names are long (e.g. movie titles like "*The Hurt Locker*", "*Slumdog Millionaire*"), vertical bar charts force labels to collide, overlap, or require severe 90° rotation that strains human neck reading.

Horizontal Bar Charts (`plt.barh`) solve this by placing categories on the **Y-axis** and numerical values on the **X-axis**, giving unlimited horizontal room for lengthy labels.

2. Syntax & Inversion of Axes

```
plt.barh(  
    y,                # Categorical labels on vertical axis  
    width,            # Length of horizontal bars (the numeric values)  
    height=0.8,       # Vertical thickness of bars  
    color='steelblue', # Color  
    edgecolor='black'  # Border outline  
)
```

[!NOTE] In `plt.bar()`, the numerical values go into `height=`. In `plt.barh()`, the numerical values go into `width=`, because bars extend horizontally!

3. Code Implementation

```
import matplotlib.pyplot as plt

oscar_movies = [
    "The Dark Knight",
    "The Hurt Locker",
    "The King's Speech",
    "The Artist",
    "Argo"
]
oscar_revenue = [1005, 170, 427, 133, 232] # in $M

plt.figure(figsize=(9, 5))
plt.barh(oscar_movies, oscar_revenue, color="teal", edgecolor="black", height=0.6)

plt.title("Box Office Revenue for Oscar Best Picture Winners", fontsize=14, fontw
plt.xlabel("Revenue (in $ Millions)", fontsize=12)
plt.ylabel("Movie Title", fontsize=12)
plt.grid(axis='x', linestyle=':', alpha=0.6)
plt.tight_layout()
plt.show()
```

4. Key Takeaways

- Use `plt.barh` whenever you have ≥ 7 categories or when category strings exceed 10 characters.
- Invert the y-axis if desired (`plt.gca().invert_yaxis()`) to display the highest ranked category at the very top.